

The Title is Not Dog Treats

By

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Date defended: MM DD, YYYY

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that this is the approved version of the following thesis/dissertation :

The Title is Not Dog Treats

Chair: Perry Alexander

Date approved: _____ MM DD, YYYY

Abstract

Formal verification is a useful tool in guaranteeing the security and behavior of programs. This is increasingly true as the amount of LLM generated code continues to rise. However even simple projects can be labor intensive to verify, thus verification see's little use in practice. Existing methods have either used program synthesis algorithms or LLM's to automate verification. Now applied blindly and on entire proofs, LLM's lack much of the resources that programmers are able to utilize and thus are not as effective in writing proofs. We present a method for introducing relevant information from the environment to the LLM, while using it to develop each proof iteratively and a tool based off this method called NotDogTreats. This consists of tracking and ranking information in the local context and providing only the most relevant information to the LLM. The process then iteratively attempts to solve each part of a proof one step at a time, allowing us to see if we are making progress. [Discuss Evaluation once Evaluated]

Acknowledgements

This is where you will insert your acknowledgements. Usually in this section, students will mention their thesis/dissertation chair and committee, any research associates, their friends, and their family members. But who you thank is ultimately up to you.

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Chapter 1

Introduction

Abstract

This is where you will insert your chapter abstract, which functions as a concise, self-contained summary of your chapter.

1.1 Overview

[What is Formal Methods and its significance?] The desire for high availability and secure software has only grown as we continue to rely on computers for critically important functions. Whether that be for banking services, pacemaker devices, or AWS a service which hosts [stupid amount of money] in business every second. For any of these cases even minor malfunctions could have significant consequences. Thus one might be interested in guaranteeing correct behavior for these systems. This is the goal of formal methods where we can formally prove properties about a program via [math...]. However formal methods has seen little use in practical applications as the process is unfamiliar to the typical programmer and time-intensive. [Reference a large formally verified program like C compiler project?]. [Introduce automated proof tools].

[Discuss past work in proof automation, then the rise of LLM based tools]

[Introduce our work in this field].

Type out the content of your subsection here. What follows is an example APA 7th citation (Metz, 1978). Note also Metz, Einstein, et al. (1978) but Metz, Hamlin, et al. (1978). You can adjust citational information by editing the allcites.bib file, included in this package.

1.2 Background?

Discuss formal methods and proof synthesis. What someone might need to know to understand this paper, How did the field develop to where we are today, and the significance of the field.

Chapter 2

Literature Review

Abstract

This is where you will insert your chapter abstract, which functions as a concise, self-contained summary of your chapter.

2.1 Name of Subsection

Type out the content of your subsection here (Einstein, 1905). Delete or duplicate as necessary.

Note that you can also refer to figures in the text like this: see figure 2.1.

Figure 2.1: Insert figure caption here



2.1.1 Name of Third-Level Subsection

Here is an example of a third-level subsection. Delete or duplicate as necessary.

2.1.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level subsection. Delete or duplicate as necessary.

Chapter 3

Methodology

Abstract

This is where you will insert your chapter abstract, which functions as a concise, self-contained summary of your chapter.

3.1 Name of Subsection

Type out the content of your subsection here. Here's another citation example (Littell et al., 1996; Metz, 1978; Wellek, 2003). And here is an example equation:

$$D = 2.91L^{0.71} \tag{3.1}$$

3.2 Name of Subsection

Here is another example of a chapter subsection. Delete or duplicate as necessary.

3.2.1 Name of Third-Level Subsection

Here is another example of a third-level subsection. Delete or duplicate as necessary.

3.2.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level subsection. Delete or duplicate as necessary.

Chapter 4

Findings

Abstract

This is where you will insert your chapter abstract, which functions as a concise, self-contained summary of your chapter.

4.1 Name of Subsection

Type out the content of your subsection here. Here's another citation (Einstein, 1905; Thomas, 1998). What follows is an example table:

Table 4.1: Insert table caption here

Set	Test 1	Test 2	Test 3	Test 4
1	50	34	837	970
2	47	877	34	230
3	5	31	25	415
4	35	27	144	2356
5	45	300	556	389

And just to be safe, here is another example of a table:

Table 4.2: Insert table caption here

Experiment	Batch A	Batch B	Batch C	Batch D	Batch E
5	7	5	3	7	2
8	9	8	2	1	9
3	2	1	9	6	6
2	2	8	1	4	9

You can also refer to tables in the text like this: see tables 4.1 and 4.2.

4.2 Name of Subsection

Here is another example of a chapter subsection. Delete or duplicate as necessary.

4.2.1 Name of Third-Level Subsection

Here is another example of a third-level subsection. Delete or duplicate as necessary.

4.2.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level subsection. Delete or duplicate as necessary.

Chapter 5

Conclusion

Abstract

This is where you will insert your chapter abstract, which functions as a concise, self-contained summary of your chapter.

5.1 Name of Subsection

Type out the content of your subsection here. Delete or duplicate as necessary.

5.2 Name of Subsection

Here is another example of a chapter subsection. Delete or duplicate as necessary.

5.2.1 Name of Third-Level Subsection

Here is another example of a third-level subsection. Delete or duplicate as necessary.

5.2.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level subsection. Delete or duplicate as necessary.

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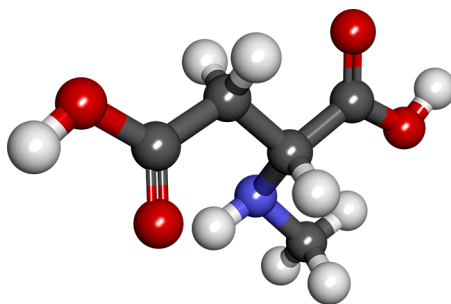
Appendix A

Title of Appendix

A.1 Name of Subsection

This is an example of an appendix. You can put citations here (Metz, 1978), as well as figures:

Figure A.1: Insert figure caption here



A.2 Name of Subsection

Here is another example of an appendix subsection. Delete or duplicate as necessary.

A.2.1 Name of Third-Level Subsection

Here is another example of a third-level appendix subsection. Delete or duplicate as necessary.

A.2.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level appendix subsection. Delete or duplicate as necessary.

Appendix B

Title of Appendix

B.1 Name of Subsection

This is an example of an appendix. Just like chapters, you can also include tables in appendices:

Table B.1: Insert table caption here

Experiment	Batch A	Batch B	Batch C	Batch D	Batch E
6	9	5	3	7	2
8	3	8	2	1	9
9	7	1	9	6	6
9	4	8	1	4	9

B.2 Name of Subsection

Here is another example of an appendix subsection. Delete or duplicate as necessary.

B.2.1 Name of Third-Level Subsection

Here is another example of a third-level appendix subsection. Delete or duplicate as necessary.

B.2.1.1 Name of Fourth-Level Subsection

Here is another example of a fourth-level appendix subsection. Delete or duplicate as necessary.